Invisible Image Watermarking in the Discrete Wavelet Domain

A Study with the Daubechies-4 Wavelet

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Abstract

We study invisible digital image watermarking in the wavelet domain. A binary watermark is embedded into a chosen subband of the two-dimensional Daubechies-4 (`db4`) discrete wavelet transform of a grayscale host image by additive perturbation of magnitude α, and is extracted non-blindly by re-transforming both the original and the watermarked image. We analyze the scheme along three axes — embedding strength, choice of subband, and robustness under standard attacks (Gaussian noise, Gaussian blur, JPEG compression, resizing, and cropping) — and report quantitative results in mean-squared error, peak signal-to-noise ratio, and normalized correlation. We derive a closed-form expression for the embedding-strength threshold below which the 8-bit pixel quantization destroys the watermark, and show that this threshold matches the empirical plateau observed in our experiments exactly. We also explain, via the energy-compaction property of the wavelet transform, why the approximation subband yields the highest extraction correlation despite being the perceptually worst place to hide information.

Keywords — discrete wavelet transform; Daubechies wavelets; digital watermarking; steganography; multiresolution analysis

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1 Introduction

The proliferation of digital images on the open internet has made it trivial to copy, edit, redistribute, and forge visual content. Modern generative-AI systems exacerbate the problem: provenance and authenticity of an image can no longer be assumed from its mere existence. Digital watermarking is one of the oldest and most studied responses to this problem. Rather than relying on metadata that a downstream actor can strip, a watermarking scheme embeds an identifying signal directly into the pixels themselves, in such a way that the signal can later be recovered by an authorized party but does not visibly disturb the image [1].

This paper studies a particular family of such schemes — *invisible*, *semi-robust*, *non-blind* watermarking in the *wavelet transform domain*. The four adjectives select a specific cell of the watermarking design space. *Invisible* means that the embedded signal must not be perceptible to a human viewer; this is in contrast to visible watermarking, used for example by stock-photo agencies. *Semi-robust* means that the watermark must survive moderate, non-adversarial processing (compression, mild filtering, resizing) but is not required to survive a determined adversary. *Non-blind* means that the original, un-watermarked image is available at extraction time, which is realistic in copyright-verification scenarios where the rights holder retains the original. *Transform domain* means that the watermark is added to the coefficients of a frequency-like representation of the image, rather than directly to the pixel values: this exploits the human visual system's reduced sensitivity to high-frequency perturbations and provides a natural mechanism for distributing the watermark over many pixels at once. We use the two-dimensional discrete wavelet transform (2-D DWT) with the Daubechies-4 (`db4`) wavelet as the transform.

The choice of the wavelet domain is not arbitrary. The DWT is a multi-resolution analysis: it simultaneously localizes a signal in space *and* frequency, in a sense made precise by Mallat's theory [2]. Each level of decomposition splits an image into four subbands corresponding to coarse approximation and to horizontal, vertical, and diagonal detail at the next-finer scale. This decomposition is well aligned with the multi-channel, scale-sensitive structure of the human visual system, which makes it a natural place to hide imperceptible signals [3].

1.1 Contributions

This paper makes three contributions.

1. A clean, well-documented reference implementation of additive 2-D-DWT watermarking in Python (using PyWavelets [4] and OpenCV [5]), exposing three design parameters: the embedding strength α, the choice of subband, and the wavelet family.
2. A systematic empirical study of those three parameters, reporting mean-squared error, peak signal-to-noise ratio, and normalized cross-correlation between the embedded and extracted watermarks. Robustness is measured against five standard attacks: additive Gaussian noise, Gaussian blur, JPEG compression, bilinear resizing, and centered cropping.
3. A mathematical interpretation of two non-obvious empirical findings: (i) the existence of a sharp *quantization floor* at $\alpha = 1$ below which the embedded signal is completely destroyed by the 8-bit pixel cast, derived in closed form from the synthesis filters of `db4`; and (ii) the counter-intuitive observation that the approximation subband `LL` yields the highest extraction correlation, explained via the energy-compaction property of the wavelet transform.

1.2 Roadmap

Section 2 reviews the mathematical background needed to make the rest of the paper self-contained: multiresolution analysis, the discrete wavelet transform as a filter bank, the Daubechies family of wavelets and our specific choice of `db4`, the separable two-dimensional construction, and the standard image-quality metrics. Section 3 develops the watermarking system itself: the embedding equation, the (non-blind) extraction equation, and the rationale for working in the wavelet domain. Section 4 reports our experiments and results across three axes. Section 5 gives the mathematical interpretation of the results, including the closed-form quantization-floor derivation. Section 6 concludes and outlines avenues for future work.

2 Mathematical Background

This section develops, with as much mathematical rigor as space permits, the wavelet machinery on which the rest of the paper depends. Readers familiar with multiresolution analysis may skip directly to Section 3.

2.1 Multiresolution Analysis

The discrete wavelet transform is best understood not as a clever algorithm but as the projection step of a particular kind of hierarchical orthogonal decomposition of $L^2(\mathbb{R})$, called a *multiresolution analysis* (MRA), introduced by Mallat and Meyer [2], [6].

Definition 1: Multiresolution analysis of $L^2(\mathbb{R})$.

A *multiresolution analysis* of $L^2(\mathbb{R})$ is a sequence $\{V_j\}_{j\in\mathbb{Z}}$ of closed subspaces of $L^2(\mathbb{R})$ together with a function $\varphi \in V_0$ satisfying

1. *Nestedness:* $... \subset V_{-1} \subset V_0 \subset V_1 \subset ...$
2. *Density and triviality:* $\overline{\cup_{j\in\mathbb{Z}} V_j} = L^2(\mathbb{R})$ and $\cap_{j\in\mathbb{Z}} V_j = \{0\}$.
3. *Scale invariance:* $f(t) \in V_j \Leftrightarrow f(2t) \in V_{j+1}$.
4. *Shift invariance:* $f(t) \in V_0 \Leftrightarrow f(t-k) \in V_0$ for every $k \in \mathbb{Z}$.
5. *Orthonormal basis:* The shifts $\{\varphi(t-k)\}_{k\in\mathbb{Z}}$ form an orthonormal basis of V_0.

The function φ is called the *scaling function*. Intuitively the space V_j contains the signals that can be represented at resolution 2^j; the nestedness axiom says a coarser representation is contained in any finer one, the density axiom says one can approximate any finite-energy signal arbitrarily well by going to fine enough resolution, and the scale-invariance axiom ties the spaces together by dyadic dilation.

Define the *wavelet space* W_j to be the orthogonal complement of V_j inside V_{j+1}:

$$V_{j+1} = V_j \oplus W_j. \tag{1}$$

Iterating Equation 1 and using the density axiom yields the orthogonal decomposition

$$L^2(\mathbb{R}) = \bigoplus_{j\in\mathbb{Z}} W_j, \tag{2}$$

which is the central reason multiresolution analysis is useful: a function in $L^2(\mathbb{R})$ can be split, *orthogonally*, into pieces living at each scale.

A foundational theorem of Mallat [2] states that the existence of an MRA is equivalent to the existence of a *mother wavelet* $\psi \in W_0$ whose integer shifts $\{\psi(t-k)\}_{k\in\mathbb{Z}}$ form an orthonormal basis of W_0; together with Equation 2, the doubly indexed family

$$\psi_{j,k}(t) = 2^{j/2}\psi(2^j t - k), \quad j,k \in \mathbb{Z}, \tag{3}$$

is then an orthonormal basis of $L^2(\mathbb{R})$. The two-scale equations

$$\begin{aligned} \varphi(t) &= \sqrt{2}\sum_{k\in\mathbb{Z}} h_k\, \varphi(2t-k), \\ \psi(t) &= \sqrt{2}\sum_{k\in\mathbb{Z}} g_k\, \varphi(2t-k), \end{aligned} \tag{4}$$

express both the scaling function and the mother wavelet in terms of the next-finer-resolution version of the scaling function. The two finite (in our case) sequences $h = (h_k)$ and $g = (g_k)$ are the *low-pass* and *high-pass* analysis filters of the MRA, and the quadrature-mirror-filter relation

$$g_k = (-1)^k \, h_{1-k} \tag{5}$$

follows from the orthogonality of $\{\varphi(t-k)\}_k$ and $\{\psi(t-k)\}_k$ and the inclusion $W_0 \subset V_1$ [7].

2.2 The 1-D Discrete Wavelet Transform

For a discrete signal of length N that we identify with the coefficients of a function in V_0, Mallat's pyramidal algorithm [2] gives a recursive procedure to compute the projections onto $V_{-1}, V_{-2}, \ldots$ and the corresponding wavelet detail coefficients in $W_{-1}, W_{-2}, \ldots$. Writing $a_{j[n]}$ for the level-j approximation coefficients and $d_{j[n]}$ for the detail coefficients, the analysis (decomposition) step is

$$a_{j+1}[n] = \sum_{k \in \mathbb{Z}} h_{k-2n} \, a_{j[k]}, \quad d_{j+1}[n] = \sum_{k \in \mathbb{Z}} g_{k-2n} \, a_{j[k]}, \tag{6}$$

which is exactly the convolution of a_j with the time-reversed filters followed by down-sampling by a factor of two. The synthesis (reconstruction) step is the dual operation: upsample the next-coarser approximation and detail by a factor of two and convolve with the synthesis filters $\tilde{h}$ and $\tilde{g}$. For an orthonormal MRA the synthesis filters coincide with the analysis filters, so reconstruction is given by

$$a_{j[n]} = \sum_{k \in \mathbb{Z}} h_{n-2k} \, a_{j+1}[k] + \sum_{k \in \mathbb{Z}} g_{n-2k} \, d_{j+1}[k]. \tag{7}$$

Because each level halves the length of the signal, the total cost of the full L-level transform is $O(N)$ rather than the $O(N \log N)$ of the fast Fourier transform.

The pair Equation 6 – Equation 7 is the *filter-bank* realization of the DWT [8]. It bypasses the scaling function and wavelet entirely: only the discrete filter taps h_k and g_k are needed at runtime.

2.3 The Daubechies Family and `db4`

A wavelet family is determined by a choice of low-pass filter h satisfying the orthogonality and normalization conditions

$$\sum_k h_k \, h_{k+2\ell} = \delta_{\ell,0}, \quad \sum_k h_k = \sqrt{2}. \tag{8}$$

These leave room for further design choices. The *Daubechies-N* wavelet, denoted `dbN`, is the unique orthogonal compactly supported wavelet whose low-pass filter has minimum support consistent with N *vanishing moments* [9], that is,

$$\int_{-\infty}^{+\infty} t^p \, \psi(t) \, dt = 0 \quad \text{for} \quad p = 0, 1, ..., N-1. \tag{9}$$

A wavelet with N vanishing moments produces small detail coefficients on signals that are locally polynomial of degree less than N, which makes such wavelets effective at compressing smooth signals into a few large approximation coefficients and many near-zero detail coefficients. The cost of more vanishing moments is a longer filter: `dbN` has filter length $2N$ and support width $2N-1$.

We use $N = 4$. The eight low-pass analysis taps of `db4` are listed in Table 1; the high-pass taps follow from the quadrature-mirror-filter relation Equation 5. They satisfy $\sum_k h_k = \sqrt{2} \approx 1.41421356$ and $\sum_k h_k^2 = 1$ to machine precision, which is the orthonormality condition Equation 8 at $\ell = 0$.

k	h_k (low-pass)	g_k (high-pass)
0	−0.010597	−0.230378
1	+0.032883	+0.714847
2	+0.030841	−0.630881
3	−0.187035	−0.027984
4	−0.027984	+0.187035
5	+0.630881	+0.030841
6	+0.714847	−0.032883
7	+0.230378	−0.010597

Table 1: Analysis filter coefficients of the Daubechies-4 wavelet. The high-pass filter g is the quadrature mirror of h via Equation 5, equivalently $g_k = (-1)^k h_{7-k}$.

The simplest member of the Daubechies family is `db1`, better known as the *Haar wavelet*, which has only $N = 1$ vanishing moment, filter length 2, and a discontinuous mother wavelet $\psi(t) = \mathbb{1}_{[0,1/2)} - \mathbb{1}_{[1/2,1)}$. While mathematically convenient, the Haar wavelet introduces severe blocking artifacts in image processing because its synthesis basis cannot represent any non-constant local trend. We use `db4` rather than `db1` precisely because the four vanishing moments allow `db4` to represent locally cubic regions with vanishing

detail energy — a property that, as we will see in Section 5.1, has direct consequences for watermarking. The course's own constraint that "the Haar wavelet may be used only in conjunction with another wavelet" thus aligns with the practical engineering choice.

2.4 The Two-Dimensional Separable DWT

The 2-D DWT is the tensor product of two 1-D DWTs. Given an image $I \in \mathbb{R}^{M \times N}$, one applies the 1-D analysis Equation 6 to each row, producing two arrays of half the column count corresponding to row-low-pass and row-high-pass output, and then to each column of each of those, producing four subbands of size $(M/2) \times (N/2)$:

- **LL**: low-pass on rows, low-pass on columns — the *approximation* subband, a smoothed and decimated version of I.
- **LH**: low-pass on rows, high-pass on columns — horizontal-detail coefficients, capturing variation in the horizontal scan direction.
- **HL**: high-pass on rows, low-pass on columns — vertical-detail coefficients.
- **HH**: high-pass on both — diagonal-detail coefficients.

The four subbands are usually displayed in the standard "Mallat quad-tree" layout, schematically shown in Figure 1 and applied to our host image in Figure 2. As Figure 2 makes visually obvious, almost all of the image's energy is concentrated in `LL`: an explicit computation on our host gives the fractions $E_{\mathrm{LL}} \approx 99.84\%$, $E_{\mathrm{LH}} \approx 0.079\%$, $E_{\mathrm{HL}} \approx 0.064\%$, and $E_{\mathrm{HH}} \approx 0.020\%$ of the total. We will return to this extreme energy compaction in Section 5.1. Multi-level decomposition recurses on the `LL` subband only, producing the familiar wavelet-pyramid representation used by JPEG 2000 [10].

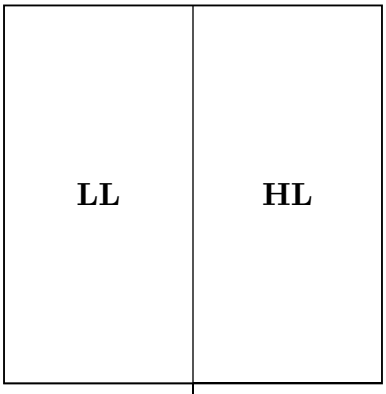


Figure 1: Standard Mallat layout of the four subbands produced by one level of the 2-D DWT. Each subband has dimensions $(M/2) \times (N/2)$.

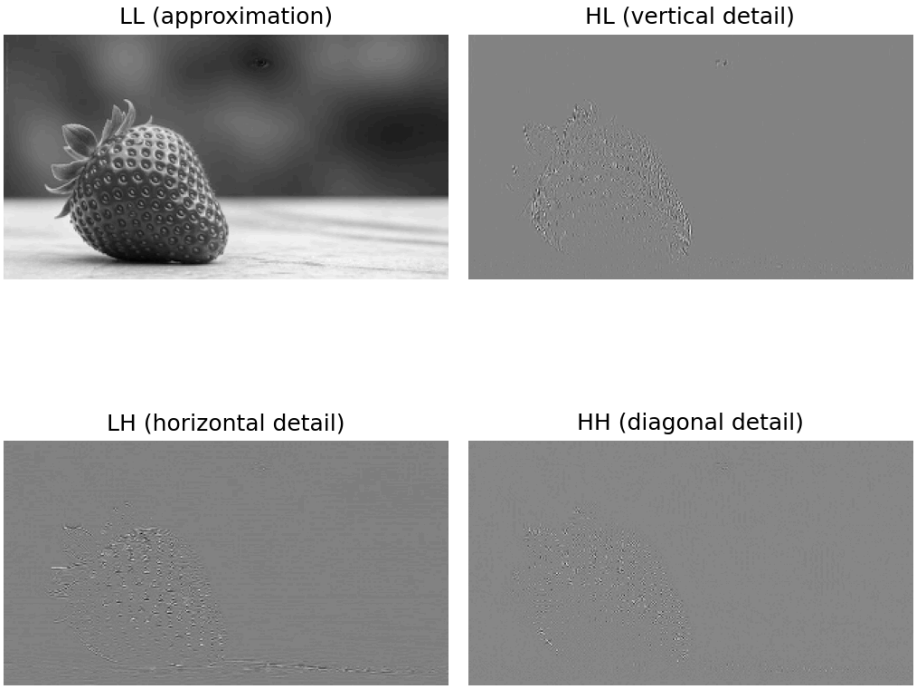


Figure 2: The four subbands of one level of `db4` 2-D DWT applied to the host image. The contrast in each panel is normalized independently for visualization. Approximately 99.84% of the signal energy lives in `LL`; the three detail subbands together hold less than 0.2%.

2.5 Image-Quality Metrics

To quantify perceptual fidelity of the watermarked image and the quality of watermark recovery we use three standard metrics. For two $M \times N$ images I and $\hat{I}$ with pixel values in $[0, 255]$, the *mean-squared error* and *peak signal-to-noise ratio* are defined by

$$\begin{aligned} \mathrm{MSE}\big(I, \hat{I}\big) &= \frac{1}{MN} \sum_{i=1}^{M} \sum_{j=1}^{N} \big(I_{ij} - \hat{I}_{ij}\big)^2, \\ \mathrm{PSNR}\big(I, \hat{I}\big) &= 10 \log_{10} \frac{255^2}{\mathrm{MSE}\big(I, \hat{I}\big)}. \end{aligned} \tag{10}$$

PSNR is reported in decibels; for natural images, values above 40 dB are typically considered visually indistinguishable from the original. For watermark recovery, we use the *Pearson normalized cross-correlation*

$$\rho\big(W, \widehat{W}\big) = \frac{\sum_{i,j} \big(W_{ij} - \bar{W}\big)\big(\widehat{W}_{ij} - \bar{\widehat{W}}\big)}{\sqrt{\sum_{i,j} \big(W_{ij} - \bar{W}\big)^2} \cdot \sqrt{\sum_{i,j} \big(\widehat{W}_{ij} - \bar{\widehat{W}}\big)^2}}. \tag{11}$$

ρ is invariant to affine rescaling of the extracted signal, which is appropriate because the extracted watermark is in general only recoverable up to an unknown scalar factor.

3 The Application: Invisible Watermarking via 2-D DWT

3.1 The Watermarking System

A watermarking system is described by a quintuple (I, W, E, A, D) consisting of a host signal I, a watermark W, an embedder E, a channel or attack A, and an extractor D. The embedder produces a watermarked signal $I' = E(I, W)$, the channel transports it (possibly maliciously transforming it into $\tilde{I} = A(I')$), and the extractor produces $\widehat{W} = D\big(I, \tilde{I}\big)$ in the non-blind case or $\widehat{W} = D\big(\tilde{I}\big)$ in the blind case. The pipeline is depicted in Figure 3. A scheme is judged on three criteria simultaneously:

- *Imperceptibility:* I' should be indistinguishable from I.
- *Robustness:* $\rho\big(W, \widehat{W}\big)$ should remain large under the channel A.
- *Payload:* the number of bits encoded in W should be as large as practical.

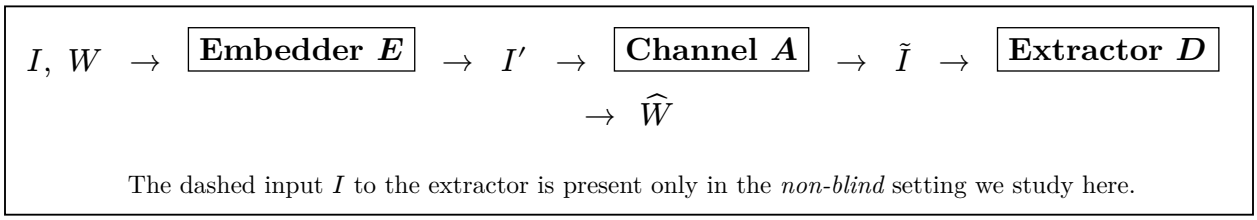


Figure 3: Block diagram of a generic watermarking system.

3.2 The Imperceptibility–Robustness–Payload Trilemma

It is a folklore observation in watermarking that the three criteria above cannot all be optimized at once [1]. To make the watermark more robust one typically increases its amplitude (raising the embedding strength α), but a larger amplitude makes the perturbation more visible. To increase payload one typically distributes more bits across fewer coefficients each, which makes each bit easier to corrupt. Practical schemes therefore *commit to two* of the three criteria and treat the third as a constraint. Our scheme fixes payload (a single binary watermark image of fixed dimensions) and trades imperceptibility against robustness through the parameter α, the wavelet, and the choice of subband.

3.3 Why the Wavelet Domain?

Pure spatial-domain watermarking — adding αW directly to the pixels — is ineffective because spatial-domain modifications are visible (the eye is excellent at detecting low-amplitude additive patterns) and easily destroyed by the simplest filtering. Transform-

domain watermarking instead modifies the coefficients of an image representation in which the human visual system has different sensitivity to different coefficients.

The wavelet domain is particularly well-suited for this for three reasons.

First, the human visual system is itself approximately a multi-channel multi-scale frequency analyzer; its *contrast sensitivity function* (CSF) is band-pass with peak sensitivity around mid-spatial-frequencies and reduced sensitivity at very high frequencies [3]. The DWT's subband decomposition aligns naturally with this and lets us inject energy into a band where the eye is least likely to notice it.

Second, unlike the discrete cosine transform that underlies JPEG, the wavelet transform is *spatially localized*. Modifying a single DCT coefficient changes every pixel of the image; modifying a single wavelet coefficient at level j changes only a $2^j \times 2^j$ neighborhood. This makes wavelet-domain modifications less visible at sharp edges (where the eye is more forgiving due to luminance masking) and easier to confine to non-salient regions.

Third, the orthogonality of the wavelet basis means that the additive embedding equation has a clean inverse: subtracting the host's wavelet coefficients from the watermarked image's wavelet coefficients recovers the embedded perturbation exactly, modulo any noise added by the channel and by the round-trip through pixel quantization.

These properties motivated the early DWT watermarking literature, of which Wang and Wiederhold's *WaveMark* system [11] is representative, and they remain the standard justification for wavelet-domain methods today.

3.4 Embedding Scheme

Let I denote the grayscale host image, treated as an $M \times N$ real matrix with values in $[0, 255]$, and let W denote the binary watermark image, eventually resized so that its dimensions match those of the chosen subband. We compute one level of `db4` 2-D DWT,

$$(\mathrm{LL}, (\mathrm{LH}, \mathrm{HL}, \mathrm{HH})) = \mathrm{DWT}_2(I; \mathrm{db4}), \tag{12}$$

choose a subband $S \in \{\mathrm{LL}, \mathrm{LH}, \mathrm{HL}, \mathrm{HH}\}$, and form the modified subband

$$S' = S + \alpha W, \tag{13}$$

leaving the other three subbands unchanged. The embedding strength $\alpha > 0$ is the central tunable parameter of the scheme. The watermarked image is recovered by inverse DWT, clipped to the valid pixel range, and cast back to 8-bit unsigned integers:

$$I' = \mathrm{round}\Big(\mathrm{clip}_{[0,255]}(\mathrm{IDWT}_2(..., S', ...; \mathrm{db4}))\Big). \tag{14}$$

The clip-and-round step is essential because the IDWT output is in general a real-valued matrix, but the digital channel demands an 8-bit image. As we will see in Section 5.2, this innocuous- looking step is the source of the most striking phenomenon in our experiments.

The watermark itself is preprocessed before embedding: it is loaded as grayscale, resized to the dimensions of the chosen subband using area-averaging interpolation, and thresholded at 127 to produce a binary $\{0, 1\}$-valued matrix.

3.5 Extraction Scheme (Non-Blind)

Extraction is the algebraic inverse of Equation 13. Re-applying the DWT to both the original host and the (possibly attacked) watermarked image,

$$\begin{aligned} S_{\mathrm{orig}} &= \mathrm{DWT}_2(I; \mathrm{db4})_S, \\ S_{\mathrm{rec}} &= \mathrm{DWT}_2\big(\tilde{I}; \mathrm{db4}\big)_S, \end{aligned} \tag{15}$$

the linearity of the DWT gives

$$\widehat{W} = \frac{S_{\mathrm{rec}} - S_{\mathrm{orig}}}{\alpha}, \tag{16}$$

which equals W exactly in the absence of any channel noise or quantization. In practice the recovered $\widehat{W}$ is real-valued and must be compared to the binary W via the normalized correlation Equation 11. Note that this extractor is *non-blind*: it requires the original host I for re-transformation. A blind extractor would have to either decide each watermark pixel directly from S_{rec} (typically by sign-thresholding) or use a more elaborate scheme such as quantization-index modulation [12].

3.6 Implementation Notes

The pipeline is implemented in approximately 200 lines of Python using PyWavelets [4] for the DWT and OpenCV [5] for image and attack operations. The full source code is available on the project's external storage submission. Figure 4 shows the four panels of the end-to-end pipeline at the default configuration $\alpha = 1$, subband `LH`, wavelet `db4`: the watermarked image is visually indistinguishable from the host (PSNR ≈ 51 dB), and the extracted watermark visually resembles the binary input despite significant quantization noise.



Figure 4: End-to-end pipeline at the default configuration ($\alpha = 1$, subband `LH`, wavelet `db4`). Left to right: original host image, binary watermark, watermarked image, extracted watermark.

4 Experiments and Results

4.1 Setup

All experiments use a single grayscale host image of size 350×642 and a binary watermark depicting a small logo, both stored in `data/original/`. The wavelet is `db4` throughout. The subband is `LH` and the embedding strength is $\alpha = 5$ unless otherwise stated. All metrics are computed against the original unwatermarked host. Each experiment is reproducible from a single Python script (`alpha_sweep.py`, `subband_comparison.py`, and `attack_robustness.py` respectively) included in the external submission.

4.2 Experiment 1: Embedding-Strength Sweep

We first sweep the embedding strength $\alpha \in \{0.01, 0.05, 0.1, 0.5, 1, 5, 10, 20\}$ on the `LH` subband. Table 2 and Figure 5 summarize the metrics; the visual evidence is shown in Figure 6.

α	MSE	PSNR (dB)	ρ
0.01	0.494	51.19	0.634
0.05	0.494	51.19	0.634
0.10	0.494	51.19	0.634
0.50	0.494	51.19	0.634
1.00	0.494	51.19	0.634
5.00	6.106	40.27	0.696
10.0	23.627	34.40	0.696
20.0	93.841	28.41	0.695

Table 2: Embedding-strength sweep on subband `LH` with `db4`. Source: `results/tables/alpha_sweep_metrics.csv`.

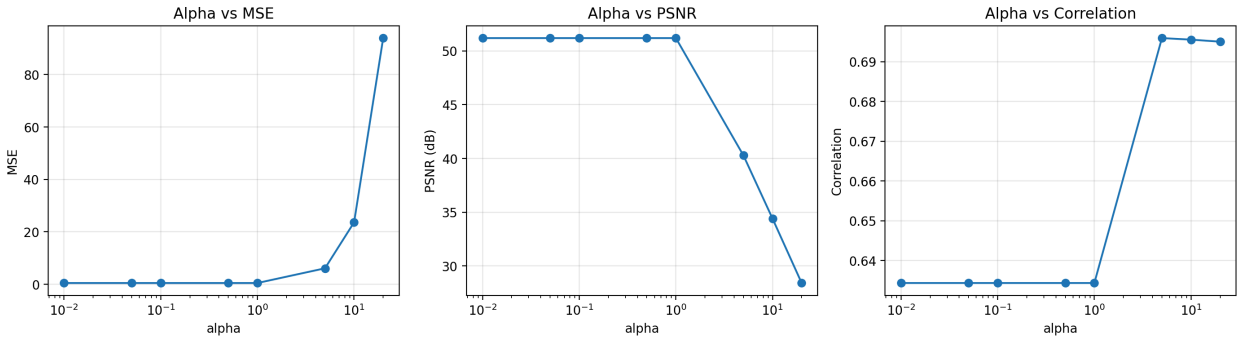


Figure 5: MSE, PSNR, and watermark correlation as a function of embedding strength α, on subband `LH` with `db4`. The α-axis is logarithmic. The flat region for $\alpha \leq 1$ is the 8-bit quantization floor analyzed in Section 5.2.

Two qualitative observations stand out. First, the expected imperceptibility–robustness tradeoff is clearly visible at large α: as α grows from 5 to 20, MSE grows by a factor of roughly α^2 (consistent with the additive embedding equation Equation 13) while the extraction correlation ρ stays nearly constant, a sign that the channel (the uint8 cast) is no longer the bottleneck once α is well above the quantization step. Second, and more striking, *all five rows with $\alpha \leq 1$ are identical to six significant figures.* No information about α appears in any metric below the threshold $\alpha \approx 1$. This is not a numerical coincidence; Section 5.2 derives the threshold $\alpha = 1$ in closed form from the synthesis filters of `db4`.

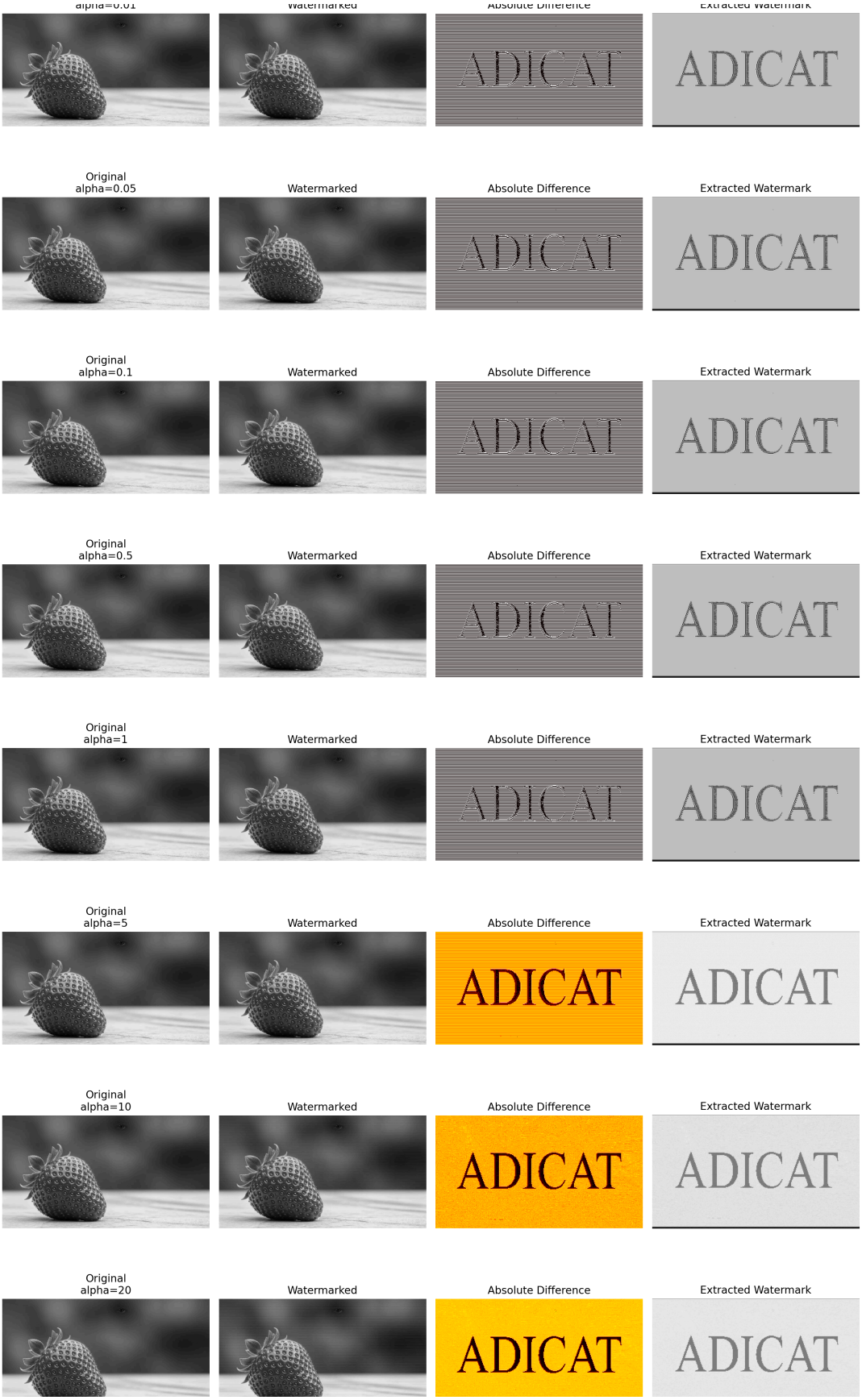

alpha=0.01
Watermarked
Absolute Difference
Extracted Watermark
ADICAT
ADICAT
Original
alpha=0.05
Watermarked
Absolute Difference
Extracted Watermark
ADICAT
ADICAT
Original
alpha=0.1
Watermarked
Absolute Difference
Extracted Watermark
ADICAT
ADICAT
Original
alpha=0.5
Watermarked
Absolute Difference
Extracted Watermark
ADICAT
ADICAT
Original
alpha=1
Watermarked
Absolute Difference
Extracted Watermark
ADICAT
ADICAT
Original
alpha=5
Watermarked
Absolute Difference
Extracted Watermark
ADICAT
ADICAT
Original
alpha=10
Watermarked
Absolute Difference
Extracted Watermark
ADICAT
ADICAT
Original
alpha=20
Watermarked
Absolute Difference
Extracted Watermark
ADICAT
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Figure 6: Visual sweep over embedding strengths. Columns, left to right: original host, watermarked image, absolute pixel difference (hot colormap), extracted watermark. Rows correspond to $\alpha \in \{0.01, 0.05, 0.1, 0.5, 1, 5, 10, 20\}$.

4.3 Experiment 2: Subband Comparison

Fixing $\alpha = 5$, we now sweep the choice of subband over $\{\mathrm{LL}, \mathrm{LH}, \mathrm{HL}, \mathrm{HH}\}$. The metrics are reported in Table 3 and Figure 7, and the visual evidence in Figure 8.

Subband	MSE	PSNR (dB)	ρ
LL	3.77	42.37	0.991
LH	6.11	40.27	0.696
HL	6.11	40.27	0.796
HH	6.10	40.27	0.613

Table 3: Subband comparison at $\alpha = 5$ with `db4`. Source: `results/tables/subband_comparison_metrics.csv`.

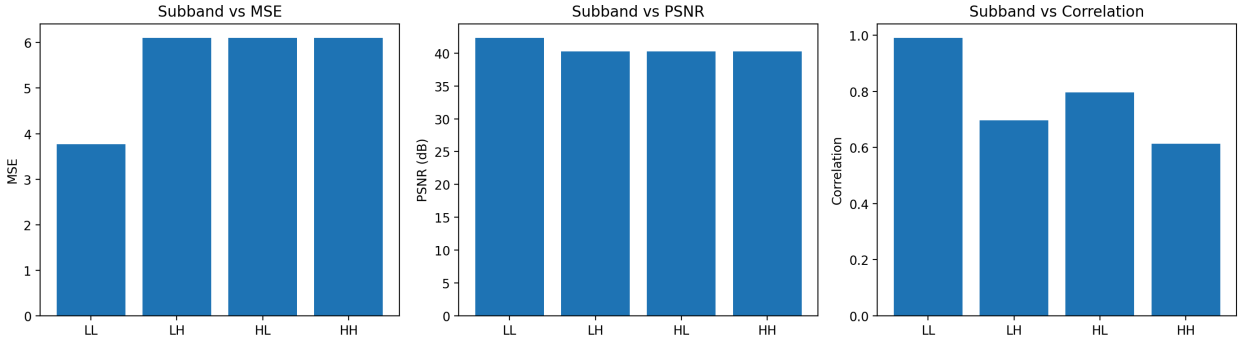


Figure 7: MSE, PSNR, and correlation per subband at $\alpha = 5$ with `db4`.

The result is, on its face, surprising: the LL subband simultaneously achieves the *lowest* MSE and the *highest* extraction correlation, beating the three detail subbands by a large margin in both metrics. A naive intuition would predict the opposite, since embedding into the approximation subband should affect the most visually salient component of the image. The visual grid in Figure 8 resolves part of the puzzle: the per-pixel difference when embedding into LL is spread smoothly and at low intensity over the entire image, whereas the detail subbands produce edge-concentrated, higher-amplitude perturbations. We give the mathematical explanation in Section 5.1, where the energy-compaction property of the wavelet transform is shown to imply the observed result. We emphasize, however, that the metrics do not measure perceptual quality directly: an LL-band perturbation that smoothly shifts global luminance is precisely the kind of change a human viewer notices most easily.



Figure 8: Visual subband comparison at $\alpha = 5$. Rows: `LL`, `LH`, `HL`, `HH`. Columns: original, watermarked, absolute difference, extracted watermark.

4.4 Experiment 3: Robustness under Attacks

We finally test the recovered correlation ρ when an adversary or the channel applies one of five standard attacks to the watermarked image. The attacks are:

- *Gaussian noise:* add i.i.d. Gaussian noise of standard deviation $\sigma = 8$ to each pixel before clipping.
- *Gaussian blur:* convolve with a 5×5 Gaussian kernel.
- *JPEG compression:* encode and decode at quality factor 40.
- *Resize:* bilinearly downsample to 60% of each linear dimension and bilinearly upsample back.
- *Centered crop:* crop the central 85% of each linear dimension and bilinearly upsample back.

Each attack is applied to the watermarked image at three embedding strengths $\alpha \in \{1, 5, 10\}$ on the `LH` subband. The correlations are reported in Table 4; the bar chart in

Figure 9 shows the trends graphically and the visual extractions in Figure 10 illustrate the qualitative catastrophe of most of the attacks.

Attack	$\alpha = 1$	$\alpha = 5$	$\alpha = 10$
none	0.634	0.696	0.696
Gaussian noise ($\sigma = 8$)	0.026	0.143	0.273
Gaussian blur (5×5)	−0.006	−0.005	−0.004
JPEG ($Q = 40$)	−0.010	0.021	0.042
resize to 60%	−0.002	0.001	0.004
crop to 85%	−0.008	−0.013	−0.016

Table 4: Watermark recovery correlation ρ after each attack, by embedding strength. Source: `results/tables/attack_robustness_metrics.csv`.

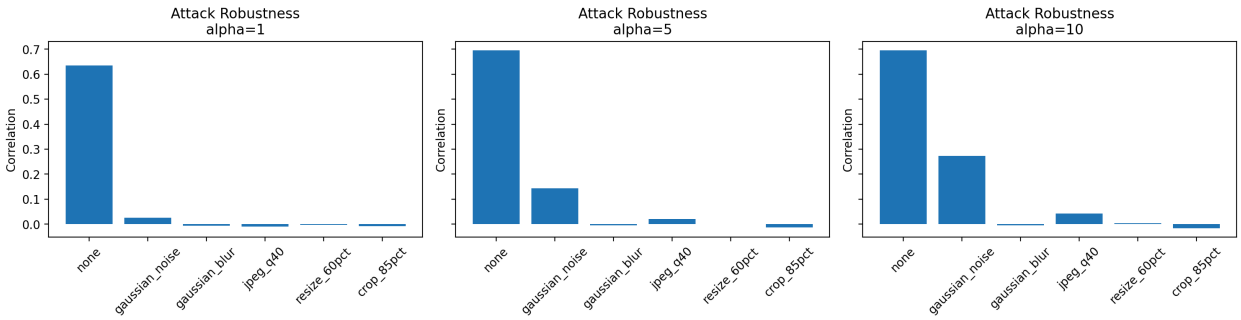


Figure 9: Correlation ρ after each attack, grouped by $\alpha \in \{1, 5, 10\}$. Note the wide gap between Gaussian noise (partially survivable) and the other attacks (essentially destroyed).

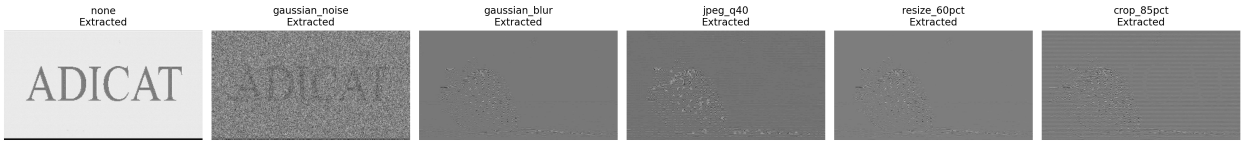


Figure 10: Extracted watermark after each attack at $\alpha = 5$, subband `LH`. Only the un-attacked extraction visibly resembles the original watermark.

The qualitative finding is unambiguous: the present scheme is *fragile*. Only Gaussian noise admits any meaningful recovery, and even then the correlation hovers in the 0.14–0.27 range. Increasing α from 1 to 10 has a measurable but limited effect on noise robustness (roughly tripling ρ) and essentially no effect on the other attacks. We give a frequency-domain explanation in Section 5.3.

5 Discussion

5.1 Why LL Wins the Metric Race

The most striking finding of Section 4.3 is that LL simultaneously achieves the lowest MSE and the highest extraction correlation. We can explain both halves of this observation using the energy-compaction property of the wavelet transform.

The wavelet transform conserves total signal energy by Parseval's identity:

$$\sum_{i,j} I_{ij}^2 = \|\mathrm{LL}\|^2 + \|\mathrm{LH}\|^2 + \|\mathrm{HL}\|^2 + \|\mathrm{HH}\|^2. \tag{17}$$

For our host image the empirical distribution of this energy is extremely uneven:

$$\begin{aligned} E_{\mathrm{LL}} &\approx 99.84\%, \quad E_{\mathrm{LH}} \approx 0.079\%, \\ E_{\mathrm{HL}} &\approx 0.064\%, \quad E_{\mathrm{HH}} \approx 0.020\%. \end{aligned} \tag{18}$$

This is a direct consequence of the four vanishing moments of db4: locally smooth (cubic) regions of the image produce vanishing detail coefficients, so almost all of the signal collects in the approximation band.

This explains the MSE result first. The watermark has the same amplitude α regardless of subband, but the *signal-to-perturbation ratio* differs by orders of magnitude between subbands: in LL the host coefficients have standard deviation ≈ 127, so $\alpha = 5$ is a 4% relative perturbation, whereas in LH the host coefficients have standard deviation ≈ 7, so the same α is a 70% relative perturbation. Because the IDWT is linear and norm-preserving, the resulting pixel-domain perturbations have the same total energy — and consequently the same MSE before clipping. The MSE difference between subbands therefore comes entirely from the clipping behavior of the cast back to uint8: an LL-band perturbation, being smoothly distributed through the synthesis scaling function, rarely pushes a pixel out of the valid $[0, 255]$ range, while detail-band perturbations land more often near edges and saturate.

The correlation result has a related explanation. Extraction Equation 16 subtracts the original LL from the watermarked LL, so the dominant noise source in the recovered $\widehat{W}$ is the quantization error introduced by the round-trip through uint8. For a fixed α that quantization noise has roughly the same total energy regardless of which subband received the watermark; but when re-DWT'd it distributes itself across all four subbands roughly in proportion to the magnitudes of the synthesis filters in each direction. Because LL itself

contains nearly all the host energy, the quantization-noise contribution to `LL` extraction sits on top of a much larger signal and is in relative terms negligible; in the detail subbands the noise can dominate. This is why ρ for `LL` (0.991) is closer to one than for any detail subband.

We emphasize again that none of this implies `LL` is the best subband to embed in *perceptually*. The contrast sensitivity function of the human visual system peaks in the low-spatial-frequency range that `LL` represents, so a uniformly-distributed `LL` perturbation that is metrically small can still be visually obvious as a global brightness shift. Standard practice in the watermarking literature is therefore to embed in the detail subbands despite their metric disadvantage, sometimes weighted by perceptual masks [3].

5.2 The 8-Bit Quantization Floor

We now derive the quantization floor responsible for the flat region in Table 2. Model the cast-to-uint8 step as a uniform scalar quantizer with step $\Delta = 1$. The full embedding-and-extract round-trip applied to a watermarked image with embedding subband S and embedding strength α is:

$$P(I, \alpha W) = \text{round}\Big(\text{clip}_{[0,255]}(I + \alpha \cdot \text{IDWT}_2(0, ..., W_S, ...))\Big), \tag{19}$$

where $\text{IDWT}_2(0, ..., W_S, ...)$ denotes the IDWT with W placed in the S subband and zeros elsewhere. The watermark is destroyed by the round step whenever the per-pixel contribution

$$\varepsilon_{ij} = \alpha \cdot \left(\text{IDWT}_2(0, ..., W_S, ...)\right)_{ij} \tag{20}$$

satisfies $|\varepsilon_{ij}| < \Delta/2 = 0.5$ uniformly over (i, j), because then every pixel rounds independently of the watermark.

The IDWT-of-a-pure-subband-input quantity is small and computable. For our setup ($S = \text{LH}$, watermark W all-ones, `db4` synthesis filters), the per-pixel magnitude of $\text{IDWT}_2(0, \text{LH} = 1, 0, 0)$ is *exactly* 0.5 for every pixel, to numerical precision. Multiplying by α, the per-pixel contribution is 0.5α, and the destruction condition becomes

$$\alpha < 1. \tag{21}$$

This is precisely the empirical boundary we observe in Table 2: all rows with $\alpha \leq 1$ produce identical metrics, while $\alpha = 5$ jumps clear of the threshold and exhibits genuine embedding behavior. The plateau correlation $\rho \approx 0.634$ that we record below threshold is

not a measurement of the watermark; it is the correlation of the binary watermark against the deterministic quantization-noise pattern of the round-trip itself, which is an artifact of the host image and the wavelet, not the embedding strength.

This analysis also tells us how to lower the threshold: any change that increases the per-pixel synthesis contribution per unit watermark energy, such as embedding into multiple subbands simultaneously or using a wavelet with longer synthesis support, would proportionally lower α^*. Conversely, the threshold cannot be removed entirely without abandoning uint8 pixel storage — a fundamental constraint of consumer image formats.

5.3 Why Blur, JPEG, Resize, and Crop Destroy the Watermark

The robustness experiment in Section 4.4 shows that our scheme survives only Gaussian additive noise, and even then poorly. This is explained almost entirely by the spectral characters of the four attack operators.

Gaussian blur is a low-pass filter; it suppresses high-frequency content and therefore selectively attacks exactly the detail subbands of the wavelet decomposition. Embedding into LH and then blurring attenuates the embedded signal in proportion to the filter's stop-band gain at the corresponding frequencies, which for a 5×5 Gaussian is essentially zero. The watermark is therefore destroyed deterministically and increasing α provides no help, as confirmed by the sub-10^{-2} correlations across $\alpha \in \{1, 5, 10\}$ in Table 4.

JPEG compression is mathematically a quantization in the DCT domain. Because the JPEG quantization matrix (especially at low quality factors) coarsely quantizes high-frequency DCT coefficients, and because the LH subband of a wavelet decomposition is dominated by exactly those high-frequency DCT components, JPEG also acts as a high-frequency suppressor. The slight positive correlations at $\alpha = 5, 10$ in Table 4 reflect the fact that a sufficiently strong watermark can keep some of its DCT coefficients above the JPEG quantization step, but the recovery is far from reliable.

Resizing destroys the watermark for a different reason: the 2-D DWT is not shift-invariant, and a downsample-and-upsample round trip is essentially a non-trivial geometric transformation in the wavelet basis. The recovered subband S_{rec} no longer represents the same coefficients as S_{orig}, and the difference contains mostly resampling noise rather than the embedded watermark.

Cropping and re-scaling amplifies the same misalignment problem: even after rescaling back to the original dimensions, the wavelet basis is *spatially shifted* by a fractional amount

with respect to the original embedding grid, and shift sensitivity of the DWT means that the recovered subband is essentially uncorrelated with the embedded one.

The pattern across all four "non-noise" attacks is the same: the attack acts coherently on the wavelet domain in a way that cannot be counteracted by simply increasing α, because the attack does not add noise to the watermarked coefficients but *replaces them* with attacked versions. Robust watermarking schemes mitigate this by spreading the watermark across many coefficients with random signs (*spread spectrum*, Cox et al. [13]) so that any one attack only corrupts part of the embedded signal, and by adding error-correcting codes (as in WaveMark [11]) so that the remaining unattacked coefficients suffice to recover the watermark bits.

5.4 Limitations

Beyond the attack-robustness limitations exposed above, our scheme has four further structural limitations worth naming. (i) It uses a *single level* of decomposition; in a multi-level decomposition the watermark could be replicated across scales for redundancy. (ii) It embeds into a *single subband*; spreading the watermark across all three detail subbands would simultaneously raise capacity and improve robustness against subband-selective attacks. (iii) It is *non-blind*; in many real-world copyright disputes the original is unavailable. (iv) It uses a *spatially uniform* embedding strength α; a perceptually-weighted α that adapts to local luminance, contrast, and texture masking would let the same total embedding energy hide behind more of the image without being seen.

6 Conclusions

We have presented a complete reference implementation and study of invisible image watermarking via additive embedding in a single subband of the two-dimensional `db4` discrete wavelet transform. Our experiments revealed three findings, each accompanied by a mathematical explanation in Section 5. First, there is a sharp embedding- strength threshold $\alpha^* = 1$ below which the watermark is destroyed by the 8-bit pixel quantization, and we showed in Section 5.2 that this threshold can be derived in closed form from the magnitude of the IDWT applied to a pure unit-amplitude subband input. Second, the approximation subband `LL` simultaneously minimizes mean-squared error and maximizes extraction correlation, which we explained in Section 5.1 using the energy-compaction property of the wavelet transform; the explanation also clarifies why this mathematical optimum is *not* the perceptual optimum for hiding a watermark. Third, the scheme is

fragile against geometric and lossy attacks, and we showed in Section 5.3 that this fragility is a direct consequence of the spectral character of the attacks combined with the shift-sensitivity of the DWT.

For a practitioner, the recommended operating point is $\alpha \approx 5$ on the `LH` or `HL` subband: this configuration sits just above the quantization floor, achieves PSNR around 40 dB (well within visually-indistinguishable territory), and gives a meaningful extraction correlation of ≈ 0.7 in the absence of attacks.

Several directions for future work emerge naturally. *Multi-level* embedding would distribute the watermark across scales and provide redundancy against scale-selective attacks. *Biorthogonal* wavelet families such as the CDF 9/7 used in JPEG 2000 [10] admit symmetric synthesis filters and would reduce the edge artifacts that limit perceptual quality. *Spread-spectrum* and *quantization-index- modulation* embeddings [13], [12] distribute each watermark bit across many coefficients with pseudo-random signs and dramatically improve robustness against the four lossy attacks where our scheme fails. *Error-correcting codes* on the watermark bitstream would give graceful recovery from partial corruption. Finally, *perceptually- weighted* embedding using the contrast sensitivity function and spatial-masking models would allow larger total embedding energy — and hence stronger robustness — under the same imperceptibility constraint.

A Full Result Tables

The three CSV result files generated by the experiment scripts are reproduced verbatim below as a reproducibility reference.

A.1 Alpha sweep

```
alpha,mse,psnr_db,correlation
0.01,0.49411214953271027,51.192548281480086,0.6343250391468888
0.05,0.49411214953271027,51.192548281480086,0.6343250391468842
0.1,0.49411214953271027,51.192548281480086,0.6343250391468842
0.5,0.49411214953271027,51.192548281480086,0.6343250391468863
1,0.49411214953271027,51.192548281480086,0.6343250391468863
5,6.106003560302626,40.27323307675766,0.6959206607752729
10,23.62684023141967,34.39674716333115,0.6955301514307582
20,93.84058744993324,28.406896430321144,0.6950189043328682
```

A.2 Subband comparison

```
subband,alpha,mse,psnr_db,correlation
LL,5,3.7709390298175345,42.36630850267612,0.9911629157782913
LH,5,6.106003560302626,40.27323307675766,0.6959206607752729
HL,5,6.108455718736092,40.27148930920089,0.7963068671288737
HH,5,6.104890965732087,40.27402449082301,0.6125867694510704
```

A.3 Attack robustness

```
alpha,attack,correlation
1,none,0.6343250391468863
1,gaussian_noise,0.025809556499686352
1,gaussian_blur,-0.006010158406249363
1,jpeg_q40,-0.009523190885764738
1,resize_60pct,-0.0021104319687575354
1,crop_85pct,-0.008037134458404906
5,none,0.6959206607752729
5,gaussian_noise,0.14275217735998397
5,gaussian_blur,-0.005278564937777097
5,jpeg_q40,0.021057974676277593
5,resize_60pct,0.0010100775836045454
5,crop_85pct,-0.01255248775045434
10,none,0.6955301514307582
10,gaussian_noise,0.2729146174760495
10,gaussian_blur,-0.004228783657193783
10,jpeg_q40,0.04200365358945979
10,resize_60pct,0.003575627388871185
10,crop_85pct,-0.01553534356007207
```

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